🎵 Spotify Playlist Web Scraping

👩‍💻 Personal Details

👤 **Name**	: Taniya
🎓 **Roll Number**	: 8524255

🏫 **Branch**	:Artificial Intelligence And Machine Learning
🏛 **College**	: JMIETI
🏢 **Company**	: CodroidHub .Pvt.Ltd
🧑‍🏫 **Instructor**	: Devasish Kumar

📌 Project Title

Spotify Playlist Web Scraping Using Python & Selenium

📖 Project Description

This project demonstrates web scraping of Spotify playlist data using Python and Selenium. Since Spotify dynamically loads its content using JavaScript, Selenium is used to automate the browser and extract playlist information successfully.

The project collects song details from a public Spotify playlist and organizes the extracted data into a structured dataset for further analysis.

Project Features:

- 🎵 Scrape Song Names

- 🎤 Extract Artist Names
- 🔗 Collect Song Links
- 📄 Store Data in a Pandas DataFrame
- 💾 Export Data to CSV File
- ⚡ Handle Dynamic Web Pages using Selenium

This project enhances practical knowledge of browser automation, dynamic web scraping, data extraction, and data handling using Python.

🛠 Python Libraries Used

- 🐍 Python
- 🌐 Selenium
- 📊 Pandas
- 🚗 WebDriver Manager
- 📒 Jupyter Notebook

📁 Extracted Data

- 🎵 Song Name

- 🎤 Artist Name
- 🔗 Song URL

🎯 Project Outcome

✓ Successfully scraped data from a Spotify public playlist.

✓ Extracted song names, artist names, and song links accurately.

✓ Stored the extracted information in a structured Pandas DataFrame.

✓ Exported the collected dataset into a CSV file for future analysis.

✓ Improved practical knowledge of Selenium, browser automation, dynamic web scraping, and Python programming.

```

In [36]:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager
import pandas as pd
import time
```
```

In [37]:

```python
# Chrome Driver
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))

# Spotify Playlist URL
url = "https://open.spotify.com/playlist/37i9dQZEVXbMDoHDwVN2tF"
driver.get(url)

# Page Load
time.sleep(8)

# Scroll to Load more songs
for i in range(5):
    driver.execute_script("window.scrollBy(0,1000)")
    time.sleep(2)

# Song Rows
rows = driver.find_elements(By.CSS_SELECTOR, 'div[data-testid="tracklist-row"]')

song_names = []
artists = []
links = []

for row in rows:
    try:
        title = row.find_element(By.CSS_SELECTOR, 'a[data-testid="internal-track-link"]')
        song_names.append(title.text)
        links.append(title.get_attribute("href"))
    except:
        song_names.append("")
        links.append("")

    try:
        artist = row.find_elements(By.TAG_NAME, "a")[1].text
        artists.append(artist)
    except:
        artists.append("")
```

In [40]:

```python
# DataFrame
df = pd.DataFrame({
    "Song Name": song_names,
    "Artist": artists,
```

```
    "Link": links
})

df
```

Out[40]:

	Song Name	Artist	Link
0	Billie Jean	Michael Jackson	https://open.spotify.com/track/7J1uxwnxfQLu4AP...
1	Beauty And A Beat	Justin Bieber	https://open.spotify.com/track/6QFCMUUq1T2Vf5s...
2	hate that i made you love me	Ariana Grande	https://open.spotify.com/track/20jbSiX29FDX4oQ...
3	Dai Dai	Shakira	https://open.spotify.com/track/0kosUz0jePvjiz4...
4	stupid song	Olivia Rodrigo	https://open.spotify.com/track/49j6SvuvWfbEKZK...
5	SWIM	BTS	https://open.spotify.com/track/68lbSrXDORS51pm...
6	Babydoll	Dominic Fike	https://open.spotify.com/track/7yNf9YjeO5JXUE3...
7	the cure	Olivia Rodrigo	https://open.spotify.com/track/55pBIZO1cqoIdeq...
8	Earrings	Malcolm Todd	https://open.spotify.com/track/0eAuGrXyGFYwur9...
9	drop dead	Olivia Rodrigo	https://open.spotify.com/track/3fRCAPMMZ8l8P9Y...
10	Beat It	Michael Jackson	https://open.spotify.com/track/3BovdzfaX4jb5KF...
11	Self Aware	Temper City	https://open.spotify.com/track/4qW3BbQAwZsrnu8...
12	The One That Got Away	Katy Perry	https://open.spotify.com/track/2009X8GyWTqyxId...
13	back to friends	sombr	https://open.spotify.com/track/7qjZnBKE73H4Oxk...
14	honeybee	Olivia Rodrigo	https://open.spotify.com/track/4kX7xkopoXZCfE1...
15	DtMF	Bad Bunny	https://open.spotify.com/track/3sK8wGT43QFpWrv...
16	Choosin' Texas	Ella Langley	https://open.spotify.com/track/7scFxt9VhL4FJwu...
17	End of Beginning	Djo	https://open.spotify.com/track/3qhlB30KknSejml...
18	Iris	The Goo Goo Dolls	https://open.spotify.com/track/4DPdJvSMB6hmrjg...
19	Man I Need	Olivia Dean	https://open.spotify.com/track/1qbmS6ep2hbBRaE...
20	Janice STFU	Drake	https://open.spotify.com/track/514joG57v4yKTsf...
21	Chicago	Michael Jackson	https://open.spotify.com/track/5BKKy9fIJL5uM9f...

	Song Name	Artist	Link
22	The Fate of Ophelia	Taylor Swift	https://open.spotify.com/track/53iuhJlwXhSER5J...
23	Life Goes On	Oliver Tree	https://open.spotify.com/track/0eu4C55hL6x29mm...

In [39]:
```
# Save CSV
df.to_csv("spotify_playlist.csv", index=False)

driver.quit()
```

🔍 Observation

- ✔️ Successfully accessed the Spotify public playlist using Selenium.
- ✔️ Dynamic content was loaded correctly after applying page wait and scrolling techniques.
- ✔️ Song names, artist names, and song links were extracted successfully.
- ✔️ The extracted data was organized into a structured Pandas DataFrame.
- ✔️ The final dataset was exported to a CSV file for future analysis and use.
- ✔️ Selenium proved to be effective for scraping JavaScript-based websites like Spotify.

✅ Conclusion

This project successfully demonstrated how to scrape data from a dynamic website like Spotify using Python and Selenium. The required playlist information, including song names, artist names, and song links, was extracted accurately and stored in a structured format using Pandas. The collected data was exported as a CSV file, making it suitable for further analysis and reporting. Overall, the project enhanced practical knowledge of Selenium, browser automation, dynamic web scraping, data handling, and Python programming.